Dresden University of Technology

Efficient sparse attention and self-supervised learned features for transformer-based cell tracking

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Research Assignment
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What we set out to learn

Question	What would count as evidence
Where does the cost of the cutoff mask actually live?	Kernel-verified dispatch; production batch B$=64$.
Can a warp manufacture association labels without annotations?	Held-out `trpL`; pre-registered rule; compute-matched control.
Which signal carries association: position and geometry, or learned features?	Coordinates-only candidates; frame-CV and LOCO.

The talk reports what changed our mind, not what the proposal hoped.

Five things we did not expect

#	Finding	Section
1	The mask already reaches a fused cuDNN kernel; the mask itself is the cost.	§4.1
2	Removing the cutoff costs accuracy in 12/12 fold–seed cells.	§4.1
3	Attention is 10.05% of the step; the norm chain is 38.7%.	§4.1
4	The warp pretext collapses to all-negative for a structural reason.	§4.2
5	Position and geometry beat learned features by 0.015 in LOCO.	§4.3/4.4

Each item is backed by a measured result; open items are named as open.

§4.1: What does the mask actually cost?

The efficiency question, tested at production batch.

The proposal's premise was only half right

Proposal: mask disables FlashAttention[1, 2]	Measured: masked SDPA reaches fused cuDNN
Verified: forced backends, per-row kernel names	Cost: mask construction and carriage

H100 / torch 2.14; $d_{head}=64$; Trackastra's real $d_{head}=40$ is also served by cuDNN.

What we benchmarked and why

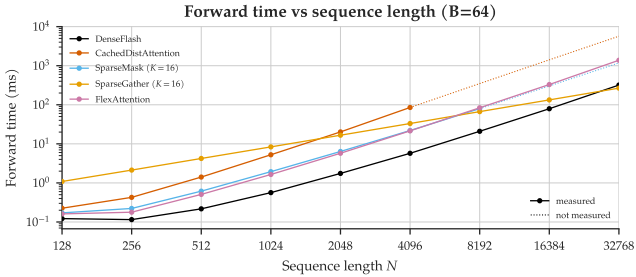
DenseFlash, CachedDistAttention, SparseGather, SparseMask, FlexAttention, MiniMax, NSA

B=64; N=128–32768; fp16; inference and training-forward

Isolated attention block; one spatial graph per (N, config); kernel names verified

B=64 is the production regime; MiniMax stays off the practical frontier.

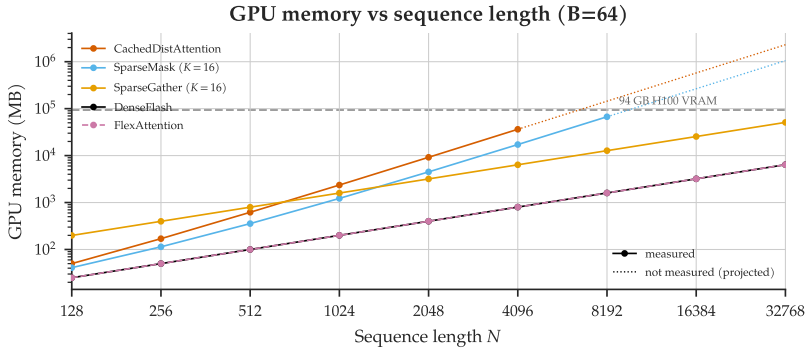
Speed at production batch: DenseFlash to 16k; SparseGather only at 32k



N	CachedDist / DenseFlash
512	6.45×
2048	11.5×
4096	14×

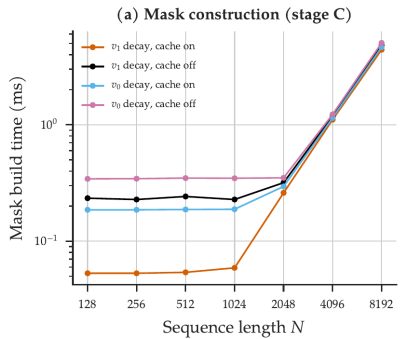
B=64; inference mode.

Memory sets the cliff, not just time



CachedDistAttention OOMs at N=8192; SparseMask at N=16384; at N=32768 SparseGather uses 50.9 GB versus 6.4 GB for FlexAttention and DenseFlash.

Where the mask cost lives: construction, not the kernel



4.41 ms

construction at N=8192

14× the kernel call

1032 MB

bool mask materialization

1.97 ms at N=8192

If you keep the cutoff, use SparseMask

N	CachedDistAttention (ms)	SparseMask K$=16$ (ms)	Speedup
512	1.407	0.615	2.3$\times$
4096	85.327	21.938	3.9$\times$

CachedDistAttention caches distances once: 1.26–1.76$\times$ over per-layer recomputation, with identical semantics.

Removing the cutoff is not accuracy-neutral

Result	Value
Worse fold–seed cells	12/12 (6 folds $\times$ 2 seeds)
Pooled AOGM penalty	$+47.5$ [32.1, 62.0]
Cutoff occupancy	42.7% mean; 23.2% median
Design	6-fold leave-one-condition-out; paired
Caveat	2 seeds; no significance test

DenseFlash (cutoff removed) versus CachedDistAttention (cutoff retained).

SparseMask inherits SparseGather accuracy

18 configurations; N≤ 512; K$\in \{4, 16\}$; 3 seeds; fp16

Forward $< \ 5 \times 10^{-4}$; backward $< \ 8 \times 10^{-4}$

$\sim$2 GPU-min versus 600+ for a full training run

Finite numerical agreement, not a proof; it licenses accuracy transfer at the tested configurations.

Token reordering does not help

Input	Reorder ratio	Verdict
Real vanvliet centroids[3]	0.992–1.023×	discarded
Random coordinates	0.97–1.03×	discarded

Proposal item; L=12; fp16; local measurement.

Attention is not the training bottleneck

Step wall

Phase	Share
norm chain	38.7%
attention	10.05%
other	51.25%

Amdahl

Assumption	Prediction
s $= 0.098$, k $= 1.75$	1.044
B$=64$, k $= 6.45$	$\approx$1.08
measured	2.9%, $\pm$6%

The norm chain is host-side: 240 device synchronizations per step; its GPU kernels are 1.6–2.1%. Dense baseline (`knn_neighbors`$= -1$); the B$=1$ k is conservative.

NSA is rejected at these N

N range	NSA / CachedDistAttention
128–1024	11.7×
2048	8.5×
4096	2.2×
8192	1.15×

Fixed compression and block-selection overhead dominates at small N, not the encoder-decoder architecture.

§4.2: Can warps manufacture association labels?

One question: can a warp supply the label for free?

Why the warp pretext is not just another augmentation

Trackastra augmentations perturb annotated windows

Warp derives $i \rightarrow T(i)^{[4,\ 5]}$

No tracking annotations; targets the low-label regime

What we tried: `trpL` held out, 5 labeled pairs, compute-matched control

`trpL` held out

5 labeled pairs

random init + compute-matched control

pre-registered success rule, fixed before the run

Result: all three arms 0.500

Arm	Validation balanced accuracy
Random init	0.500
Warp pretext	0.500
Matched control	0.500

Chance is 0.500; the pre-registered success rule was not met.

Why it collapsed: $\sim 1/\mathrm{N}$ positives and an all-negative optimum

$\sim$ $1/\mathrm{N}$ positives in the N$\times$N BCE

all-negative is the analytic optimum

loss 0.161 at the entropy floor; control pool 0.10 vs 0.188 floor

Input audit: targets match the true correspondence; view-2 appearance is correctly warped; identical code learns on another pool.

Coordinate solvability: 0.982 of warp labels recoverable from coordinates

Configuration	ICP	Warp-family	Oracle
Applied: affine $10°$, jitter 4, dropout 0.1	0.854	0.982	0.999
Wider: affine $25°$, jitter 8, dropout 0.1	0.629	0.900	0.933

Random baseline ≈ 0.03 (all columns)

This is a property of the labels, not a verdict on the pretext's training value.

What we did instead: DynaCLR encoder 1.47M, frozen

DynaCLR [6]; Achard protocol [7].



1.47M parameters versus SAM2.1's 80.8M.

This is a feature encoder, not the proposed association-head pretraining.

Deprioritized, not disproven; two branches remain open

Decision	Still open
Deprioritized as a scope decision	Distortion families
Not a measured rejection	Low-label vs full-label sweep
No full-scale run	Only one applied configuration

§4.3/§4.4: What was actually trained? What do features add?

Frozen injection and the feature probe.

What was actually trained: frozen injection, not end-to-end fine-tuning

End-to-end fine-tune: left open

Frozen DynaCLR + projection head

Association head trained on annotations

KNN probe: K=8 covers 99.3%, candidates by centroid, shared MLP

K=8 covers 99.3%

Candidates by centroid; feature never selects

Shared MLP, 3 seeds, frame-CV + LOCO

The feature cannot choose its own shortlist.

Position and geometry dominate: DynaCLR 0.826, nearest-centroid 0.811

K=8 candidates, 3 probe seeds		
Feature	Frame-CV	LOCO
Regionprops 13D	0.8694	0.8183
Hu 20D	0.8679	0.8023
DynaCLR[6]	0.8527	0.8259
Regionprops 7D	0.8422	0.7915
position-only	0.8127	0.7887
SAM2.1[8]	0.7354	0.7256
DINOv2[9]	0.6782	0.6433
NT-Xent	0.5778	0.5634
nearest-centroid	0.812	0.811

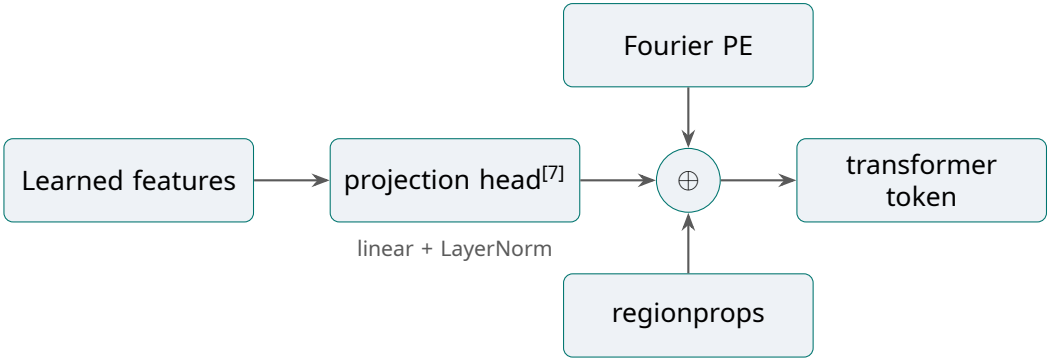
What the probe means: features add little over geometry

Features add little over geometry

SAM2.1 mid-pack

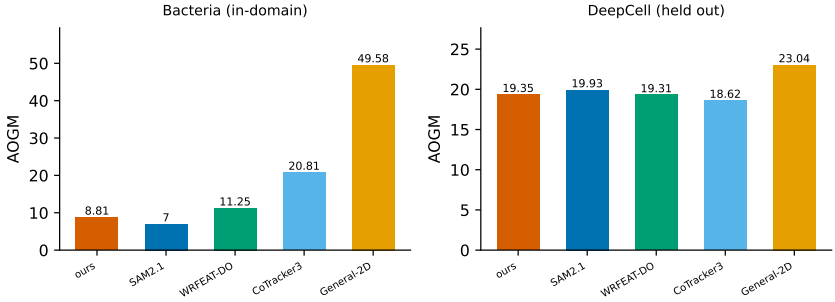
NT-Xent 0.563

Injection protocol: projection head + concat; encoder frozen



Small encoder is competitive, not state of the art

Reference protocol [7]; SAM2.1 [8]; Bacteria [10]; DeepCell [11].



Bacteria 8.81 vs SAM2.1 7.00; DeepCell 19.35 vs 19.93; Original Trackastra 11.9 (greedy). The remaining DeepCell gap is detection.

The head uses features, but not their identity

Feature condition	Bacteria AOGM
normal	8.81
shuffled	9.75
zeroed	150.6

Zeroing breaks the head; shuffling identity costs 0.94 AOGM.

Raw embedding geometry near-uniform: cosine-only fails ($\sim$88332)

AOGM ≈ 88332

cosine near-uniform

0.5 threshold over-links

Cosine-only protocol fails for this encoder; not a standalone feature-quality verdict.

§4.4 interpretation: small encoder competitive; position is the signal

1.47M competitive

Position is the signal

Not state of the art

What changed, and how we caught it

Changed belief	How we caught it
The mask cost is not a kernel-package cost.	Forced-backend arms and per-row kernel names; masked SDPA reaches fused cuDNN.
The cutoff is not only a compute knob.	Paired 6-fold leave-one-condition-out ablation: 12/12 fold–seed cells worse.
The warp pretext does not transfer at prototype scale.	Pre-registered low-label pilot failed; loss at the entropy floor; input audit ruled out a bug.
Learned features are not the main association signal.	Coordinates-only candidates; DynaCLR LOCO 0.826 vs nearest-centroid 0.811.
The delivered SSL component is a feature encoder.	Frozen DynaCLR injection; the proposed association-head pretraining was not delivered.

What we did not do, and why

Not done	Why
Full Trackastra port	Optional in the proposal; not attempted. The 18-configuration test checks SparseGather/SparseMask equivalence, not a port.
Token reordering	0.992–1.023$\times$ on real centroids; discarded.
NSA	11.7$\times$ slower than CachedDistAttention at N$=128$–1024; fixed overhead dominates at small N.
Distortion-family and low-label sweep	Open beyond the prototype; only affine 10° + jitter 4 + dropout 0.1 was piloted.

Recommendations for the Trackastra team

Recommendation	Evidence
Keep the cutoff	Removing it is worse in 12/12 fold–seed cells; pooled $+47.5$ AOGM [32.1, 62.0].
Set K$=16$	In-sample 201.8 vs 200.8 AOGM; held-out `trpL` 51.0 vs 43.3. K$=32$ is the accuracy-first alternative.
Use SparseMask	2.3$\times$ at N$=512$ and 3.9$\times$ at N$=4096$ versus CachedDistAttention.
Do not optimize attention first	10.05% of the step; Amdahl 1.044 versus 2.9% measured.
Normalize the chain	38.7% of the step and 240 synchronizations per step: the real target.

The cutoff ablation is, to our knowledge, the first for Trackastra.

What I would do next

#	Next step	What changes
1	HOCT edge-centric[12]	Replace node-level token features with edge-centric HOCT edge tokens.
2	End-to-end encoder	Unfreeze and fine-tune the encoder; the current protocol is frozen injection.
3	Linker ablation	Separate association-head quality from the linker effect on the metric.
4	Multi-seed / HP	Widen beyond the two CV seeds and add a hyperparameter search before treating small gaps as stable.

What these results do not license

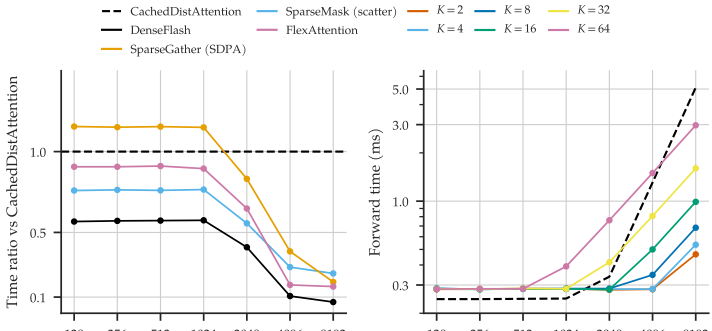
Not licensed	Why
No state-of-the-art claim	Bacteria 8.81 vs SAM2.1 7.00; DeepCell 19.35 vs 19.93; still behind Original Trackastra 11.9.
No large-N accuracy	The operator benchmark reaches N$=32768$; tracking accuracy at thousands of cells is untested.
No association-head pretraining	The delivered component is a frozen DynaCLR encoder; the warp pretext was deprioritized.
No low-label gain	Pilot arms all 0.500; the pre-registered success rule was not met.
No significance test	Two seeds, 6-fold CV; the 12 fold–seed cells share training data.

One encoder, one crop protocol; distortion families were not swept.

Thank you

Questions & discussion

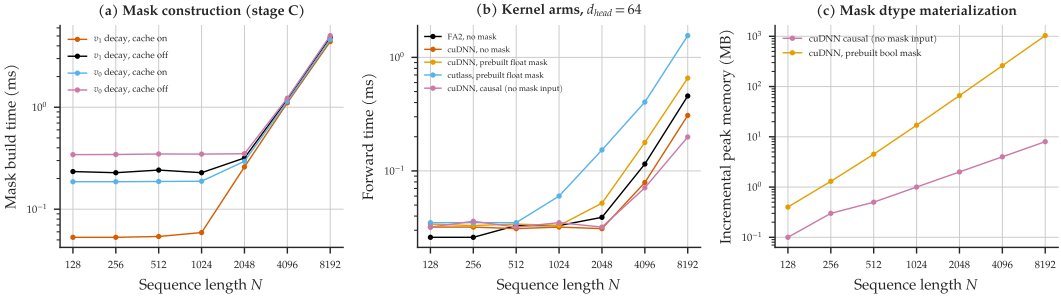
Full speed detail



(a) All architectures vs CachedDistAttention

(b) SparseGather (SDPA): context size K

Full mask decomposition



Cutoff occupancy

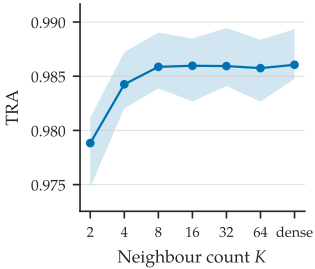
Condition	Masked fraction
cib	25.5%
metA	38.5%
pheA	42.9%
recA	50.3%
rpsM	54.4%
trpL	38.6%
Mean	42.7%
Median	23.2%

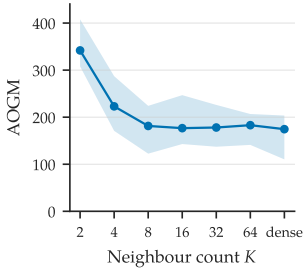
Kernel verification and equivalence

Architecture	Benchmark key and setting	Verified kernel
RelativePositionalAttention	`dense_masked` (cache off)	cuDNN
CachedDistAttention	`dense_masked` (cache on), `knn`$= -1$	cuDNN
DenseFlash	`dense_flash`, `knn`$= -2$	flash
SparseGather	`gather_matmul`, `knn`> 0	flash/cuBLAS
SparseMask	`mask_knn`, `knn`> 0	cuDNN

Measured on H100 / torch 2.14 at $d_{head}=64$; Trackastra's real $d_{head}=40$ is also served by cuDNN.

K-sweep and K=16





K	In-sample	Held-out
Dense	200.8	43.3
16	201.8	51.0
8	203.0	73.0
32	204.9	43.9

Probe table

Feature	Frame-CV top-1	LOCO top-1	LOCO F1
CNN DynaCLR (frozen)	0.8527 ± 0.0044	0.8259 ± 0.0054	0.8279 ± 0.0053
Regionprops 13D	0.8694 ± 0.0008	0.8183 ± 0.0035	0.8174 ± 0.0024
Hu 20D	0.8679 ± 0.0014	0.8023 ± 0.0052	0.8115 ± 0.0040
Regionprops 7D	0.8422 ± 0.0046	0.7915 ± 0.0020	0.7898 ± 0.0024
Position-only	0.8127 ± 0.0013	0.7887 ± 0.0040	0.7743 ± 0.0033
SAM2.1-Hiera-B+ (frozen)	0.7354 ± 0.0026	0.7256 ± 0.0044	0.7192 ± 0.0036
DINOv2 (frozen)	0.6782 ± 0.0042	0.6433 ± 0.0043	0.6520 ± 0.0031
CNN NT-Xent (frozen)	0.5778 ± 0.0037	0.5634 ± 0.0004	0.5604 ± 0.0011
Nearest-centroid (K=1)	0.812	0.811	n/a

Injection table

Method	Bacteria AOGM	DeepCell AOGM	Exposure
General-2D (Trackastra)	49.58	23.04	in-domain
WRFEAT-DO	11.25 ± 3.67	19.31 ± 0.55	in-domain
SAM2.1-Hiera-B+	7.00 ± 2.14	19.93 ± 1.15	zero-shot
CoTracker3	20.81 ± 5.41	18.62 ± 0.99	zero-shot
Original Trackastra (greedy)	29.0	11.9	in-domain; GT detections
DynaCLR ScaledCNN (ours)	8.81 ± 2.11	19.35 ± 0.76	Bacteria in-domain; DeepCell test held out (91/27)

Pretext and coordinate solvability

Configuration	ICP	Warp-family solver	Oracle	Random
Applied (affine 10° + jitter 4 + dropout 0.1)	0.854	0.982	0.999	0.032
Wider (affine 25° + jitter 8 + dropout 0.1)	0.629	0.900	0.933	0.031

Collapse audit	Value
Random init / warp pretext / matched control	0.500 / 0.500 / 0.500
Pretext SSL loss	0.161 (entropy floor of a constant predictor)
Matched control pool loss	0.10 vs 0.188 floor

Amdahl and complexity

Epoch / step share	Value
Batch phase	97.7%
Dataloader	0.5%
Validation	1.1%
Residual overhead	$\sim$0.7%
Normalization chain	38.7%
Attention	10.05%
GPU kernels	$\sim$48%
Norm GPU kernels	1.6–2.1%
Amdahl share s	0.098
Predicted, k ≈ 1.75	1.044
Predicted, B$=64$ ratio	$\approx$1.08
Measured per-epoch	2.9%
Three-seed CV spread	$\pm$6%

Architecture	Compute	Memory	Loc.
RelativePositionalAttention	$\mathcal{O}(N^2 d)$	$\mathcal{O}(N^2)$	yes
CachedDistAttention	$\mathcal{O}(N^2 d)$	$\mathcal{O}(N^2)$	yes
DenseFlash	$\mathcal{O}(N^2 d)$	$\mathcal{O}(N)$	no
SparseGather	$\mathcal{O}(Nkd)$	$\mathcal{O}(Nk)$	yes
SparseMask	$\mathcal{O}(Nkd)$	$\mathcal{O}(N^2)$	yes

References

[1] Tri Dao et al. “FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness”. In: Advances in Neural Information Processing Systems. Vol. 35. 2022. DOI: 10.52202/068431-1189.

[2] Tri Dao. “FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning”. In: International Conference on Learning Representations (ICLR). Poster. 2024. URL: https://openreview.net/forum?id=mZn2Xyh9Ec.

[3] Xiaoyang Wu et al. “Point Transformer V3: Simpler, Faster, Stronger”. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2024, pp. 4840–4851.

[4] Xiaolong Wang, Allan Jabri, and Alexei A. Efros. “Learning Correspondence From the Cycle-Consistency of Time”. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2019, pp. 2561–2571. DOI: 10.1109/CVPR.2019.00267.

[5] Allan Jabri, Andrew Owens, and Alexei A. Efros. “Space-Time Correspondence as a Contrastive Random Walk”. In: Advances in Neural Information Processing Systems. NeurIPS 2020. 2020. URL: https://arxiv.org/abs/2006.14613.

[6] Eduardo Hirata-Miyasaki et al. “DynaCLR: Contrastive Learning of Cellular Dynamics with Temporal Regularization”. In: arXiv preprint arXiv:2410.11281 (2025). DOI: 10.48550/arXiv.2410.11281.

[7] Cyril Achard. “Leveraging learned object features for Transformer-based cell tracking in microscopy”. Supervisor: Martin Weigert (ScaDS.AI, TU Dresden). Master Thesis. École Polytechnique Fédérale de Lausanne, Aug. 15, 2025.

[8] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, et al. “SAM 2: Segment Anything in Images and Videos”. In: International Conference on Learning Representations (ICLR). Oral; arXiv:2408.00714. 2025.

[9] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. “DINOv2: Learning Robust Visual Features without Supervision”. In: Transactions on Machine Learning Research (TMLR) (2024). arXiv:2304.07193.

[10] Simon van Vliet et al. “Spatially Correlated Gene Expression in Bacterial Groups: The Role of Lineage History, Spatial Gradients, and Cell-Cell Interactions”. In: Cell Systems 6.4 (2018), 496–507.e6. DOI: 10.1016/j.cels.2018.03.009.

[11] Morgan Sarah Schwartz et al. “Caliban: Accurate cell tracking and lineage construction in live-cell imaging experiments with deep learning”. In: bioRxiv (2024). DOI: 10.1101/803205. URL: https://www.biorxiv.org/content/10.1101/803205.

[12] Jordão Bragantini, Ilan Theodoro, and Loïc A Royer. “Higher-Order Cell Tracking Transformer”. In: arXiv preprint arXiv:2607.11754 (2026).